

Rajat Keshari

Software Developer and NIT Student

Resourceful **Software Developer** with expertise in designing robust, scalable system architectures and crafting efficient solutions that prioritize performance, reliability, and long-term maintainability. Proven track record of optimizing frontend and backend systems, reducing load times by up to 40% through effective state management, API optimization, and performance-driven engineering practices

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🐙 github.com/samm-developer

WORK EXPERIENCE

Software Developer as Owner

The Kings Library 📄

04/2023 - 07/2025

<http://thekingslibrary.in/>

Achievements/Tasks

- Developed full stack **Library Management System** from **scratch** with **React.js, Javascript, TypeScript, Express.js, Node.js and MongoDB**.
- Designed and implemented a **real time automatic seat allocation module** based on time slot using **transaction and rollback** that prevents booking conflicts and efficiently manages occupancy.
- Implemented secure login using **access, refresh tokens and used cache** for blocking used tokens after logout.
- Developed a **book management module** allowing users to **search, rent and purchase books** from library with real time inventory updates.
- Built **search suggestion using debouncing, throttling and redux-toolkit**, improving response latency by **60%**.
- Built **responsive and user friendly interfaces using React.js and Redux Toolkit**, enabling efficient state management and delivering a seamless user experience across desktop and mobile devices.
- Integrated 2 **secure payment gateway** RazorPay(earlier), PhonePe to enable online payments for seat reservations, book rentals, purchases.
- Integrated a **RabbitMQ-based notification microservice** for asynchronous processing of payment confirmation, fee due reminder, and booking notification emails.
- Developed RESTful APIs using Node.js and Express.js** for user management, seat reservations, book rentals/purchases, payment processing, and notification workflows.
- Optimized **MongoDB query performance by implementing indexes** on frequently queried fields, improving data retrieval speed and reducing response times for high traffic operations.

EDUCATION

M. Tech (Computer Science)

NIT Jalandhar

07/2025 - Present

74%

Courses

- Machine Learning, Advance Computer Network, Data Structure and Analysis, Operating System, Cryptography

B. Tech (Computer Science)

B.I.E.T Jhansi

07/2019 - 04/2023

74%

Courses

- Web Development, DBMS, Operating System, Compiler Design, Computer Network, OOPS

SKILLS

React.js

Next.js

JavaScript

TypeScript

MERN

AWS

MongoDB

GraphQL

Redux Toolkit

RabbitMQ

Tailwind CSS

Razorpay, PhonePe Payment Gateway

SOCKET.IO

ACHIEVEMENTS

DSA

Solved 500+ DSA problems on various platform i.e Codechef, GFG, LeetCode, strengthened logic building and algorithmic thinking through consistent practice

Portfolio

Build my own [portfolio website](#) which is inspired by Windows Operating System.

PERSONAL PROJECTS

SyncStream – Real Time Collaborative Video Platform 📄

- Built synchronized YouTube playback for multiple users using Socket.io, Node.js, and React.
- Implemented timestamp-based latency compensation achieving sub-second synchronization accuracy.
- Designed scalable client-server architecture with real-time state synchronization.
- Integrated YouTube IFrame API with robust URL parsing and edge-case handling.

YouTube Clone – Optimise search and Video Streaming 📄

- Developed video streaming platform using React and Redux Toolkit.
- Implemented optimized search with debouncing and caching, reducing API calls by 98%.
- Built nested comments using recursive components and live chat simulation.
- Integrated YouTube Data API v3 and Google Suggest API with CORS handling.
- Designed responsive UI using Tailwind CSS and React Router.

CERTIFICATES

Cyber Job Simulation Deloitte 📄

Graph Theory Programming Camp 📄

Introduction to C++ 📄

Data Structures in C++ 📄

C Language - LearnVern[CAN 13420026996257] 📄